

The Deterministic Horizon: When Extended Reasoning Fails and Tool Delegation Becomes Necessary

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Abstract

Extended chain-of-thought reasoning can degrade performance on deterministic state-tracking tasks—not due to preference biases, but fundamental information-theoretic limits in decoder-only transformers.

We establish: (1) an Attention Bottleneck Theorem with matching lower bound, proving state-tracking capacity scales as $O(H \cdot \log(L/H) \cdot \sqrt{d_h})$; (2) a context-dependent error model yielding super-exponential accuracy decay; (3) the State-Space Jaccard metric distinguishing capability from preference failures; (4) a Deterministic Horizon $d^* \in [19, 31]$ beyond which tool delegation becomes necessary.

Across 12 models and 8 task domains—including SWE-Bench, WebArena, and SQL-Multi—tool-integrated reasoning achieves 94% accuracy versus 28–42% for neural chain-of-thought. Fine-tuning on optimal-length traces yields <5% improvement, confirming an architectural ceiling. High cross-model correlation ($r = 0.81\text{--}0.91$) demonstrates these failures are architectural, not training-specific. Our results provide principled guidance for when pure neural reasoning should yield to hybrid approaches in agentic systems.

1. Introduction

The dominant paradigm in large language model reasoning posits that extended deliberation improves accuracy. OpenAI’s o1 (OpenAI, 2024), DeepSeek-R1 (Guo et al., 2025), and similar architectures invest heavily in chain-of-thought (CoT) generation, with the implicit thesis that additional inference-time compute yields better reasoning (Snell et al., 2025; Brown et al., 2025). Recent work shows test-

time compute can be more effective than parameter scaling, with compute-optimal strategies improving efficiency by $4\times$ (Snell et al., 2025; Li et al., 2024a). However, Balachandran et al. (2025) demonstrate improvements diminish with problem difficulty, and Sui et al. (2025) provide a comprehensive survey documenting the “overthinking” phenomenon.

We challenge this thesis for **deterministic state-space search**—tasks requiring exact sequences of operations transforming initial state σ_0 to target τ through finite operator set $\mathcal{O}$. Such problems pervade software engineering, formal verification, and sequential planning (Dziri et al., 2024; Kambhampati et al., 2024; Valmeekam et al., 2024). In these domains, correctness is binary; approximation is failure. Unlike open-ended generation where “mostly correct” may suffice, state-space search demands exact tracking—a requirement that exposes fundamental limitations of decoder-only attention.

The Core Phenomenon. On permutation puzzles solvable by BFS in $<0.1s$, state-of-the-art reasoning models fail after *minutes* of deliberation. Analysis of 2,847 failed traces reveals **State-Space Decoherence**: accumulated errors causing complete divergence from ground truth. This failure is *caused* by extended reasoning, not mitigated by it. The pattern is striking: at depth 10, models maintain 78% accuracy; by depth 30, accuracy drops to 19%; beyond depth 50, performance approaches random guessing (Figure 1).

Two Competing Effects. Following the framing of Kim et al. (2025), we identify tension between:

- **Complexity at reasoning time:** Deeper CoT increases cumulative error probability through attention entropy dispersion (Gong & Zhang, 2024; Barbero et al., 2024). Each step degrades signal-to-noise ratio in the residual stream.
- **Flexibility at tool time:** External computation sidesteps hard subproblems entirely (Gao et al., 2023; Gou et al., 2024; Pan et al., 2024). Tools provide exact computation without attention-based state tracking.

Distinguishing from Prior Work. Wu et al. (2025) doc-

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Table 1. Divergent predictions distinguishing Simplicity Bias (Wu et al., 2025) from Decoherence (this work). Our predictions validated (✓).

Prediction	Wu et al.	Ours	Result
Fine-tuning recovery	>30%	<5%	3.2% ✓
Length prompt gain	>10%	<2%	1.1% ✓
Cross-model r	Low	High	0.86 ✓
Enc-dec advance	None	2–3×	2.8× ✓

ument the inverted-U curve, attributing it to “Simplicity Bias”—preference for shorter reasoning. Their framework predicts training interventions can recover performance. We propose a *complementary architectural diagnosis*: even models that *attempt* long reasoning cannot maintain accuracy because autoregressive attention lacks substrate for exact state tracking. This is the **Simulator Fallacy** (Bender & Koller, 2020): conflating token prediction with algorithm execution.

Divergent Predictions. The frameworks make testable predictions (Table 1): (i) Wu et al. predict fine-tuning on optimal-length traces yields >30% recovery; we predict <5% due to architectural ceiling. (ii) Wu et al. predict prompt-level length encouragement yields >10% gains; we observe <2%. (iii) Wu et al. predict low cross-model correlation (training-specific); we observe $r > 0.8$ (architectural).

Contributions. We make six contributions:

1. **Attention Bottleneck Theorem** with information-theoretic derivation and matching lower bound, establishing capacity $O(H \cdot \log(L/H) \cdot \sqrt{d_h})$ (Section 4).
2. **Context-dependent error model** $\epsilon(d) = \epsilon_0 + \gamma d/L$ derived from attention entropy, yielding super-exponential accuracy decay (Section 4).
3. **Deterministic Horizon** d^* with closed-form formula, validated via architecture ablations ($d^* \propto \sqrt{d_h \cdot H}$) and context-length scaling ($d^* \propto \sqrt{L}$) (Section 4).
4. **SSJ metric** with precision/recall decomposition distinguishing capability (both decay) from preference (only recall decays) failure (Section 3).
5. **Empirical validation** across 12 models, 8 task domains including real-world benchmarks (SWE-Bench, WebArena, SQL-Multi), with cross-architecture comparison (Section 5).
6. **Fine-tuning experiment** confirming architectural ceiling prediction, providing key discrimination from preference-based explanations (Section 5.3).

2. Related Work

Our work synthesizes five research streams; a detailed positioning matrix appears in Appendix A.

CoT Foundations. Chain-of-thought prompting was established by Wei et al. (2022), extended by zero-shot CoT (Kojima et al., 2022), self-consistency (Wang et al., 2023), Tree-of-Thoughts (Yao et al., 2024), and Graph-of-Thoughts (Besta et al., 2024). Zelikman et al. (2022) showed bootstrapping through STaR training. Nye et al. (2021) introduced scratchpads for intermediate computation. Li et al. (2024b) proved CoT expands expressivity from AC^0 to polynomial-time; Feng et al. (2024) characterized attention pattern expressiveness. ReAct (Yao et al., 2023) synergizes reasoning with acting.

Overthinking Literature. Wu et al. (2025) provide the closest concurrent work, documenting inverted-U curves attributed to Simplicity Bias. Chen et al. (2024) examine overthinking in o1-like models. Wang et al. (2025) identify premature path abandonment. Sui et al. (2025) survey efficient reasoning. Marjanović et al. (2025) analyze DeepSeek-R1’s “sweet spot.” Su et al. (2025) study the overthinking-underthinking spectrum.

Working Memory and Over-Squashing. Gong & Zhang (2024) demonstrate attention entropy limits working memory. Barbero et al. (2024) prove representational collapse from causal masking. Liu et al. (2024) document “lost in the middle.” Xiao et al. (2024) identify attention sinks. Zhang et al. (2024) extend entropy analysis. Aksenov et al. (2025) find representation collapse. Levy et al. (2024) show length-dependent degradation. Olsson et al. (2022) discover induction heads. Elhage et al. (2021) provide circuit frameworks. Bietti et al. (2024) analyze memory from birth.

Theoretical Foundations. Merrill & Sabharwal (2024) establish expressivity bounds. Merrill et al. (2024) show SSMS share TC^0 limits. Bavandpour et al. (2025) provide CoT step lower bounds. Peng et al. (2024) prove composition impossibility via communication complexity. Hahn (2020) identify formal language limitations. Déletang et al. (2023) study Chomsky hierarchy relations. Pérez et al. (2021) prove conditional Turing completeness. Yun et al. (2020) establish universal approximation. Bhattamishra et al. (2020) analyze attention mechanism power. Strobl et al. (2024) provide formal language perspective. Information-theoretic foundations include Tishby & Zaslavsky (2015) on bottlenecks, Shwartz-Ziv & Tishby (2017) on DNN information dynamics, Goldfeld & Polyanskiy (2020) on information bottleneck framework, and Kirsch et al. (2024) connecting transformers to information theory.

Tool Augmentation. Gao et al. (2023) introduced program-aided language models. Li et al. (2023) propose Chain-of-Code. Chen et al. (2023) introduce Program of Thoughts. Gou et al. (2024) achieve SOTA via tool integration. Pan et al. (2024) demonstrate symbolic solver delegation. Schick et al. (2023) enable self-supervised tool learning. Parisi et al. (2022) introduce tool-augmented language models. Yang et al. (2024) apply RL for tool-augmented math. Qin et al. (2024) survey tool learning. Mialon et al. (2023) survey augmented LMs. Patil et al. (2023) connect LLMs to APIs.

Compositional Reasoning. Dziri et al. (2024) show transformers solve compositional tasks via linearized matching. Petty et al. (2024) demonstrate depth provides diminishing returns for compositionality. Benchmarks include Lake & Baroni (2018), SCAN/CFQ (Keysers et al., 2020), and COGS (Kim & Linzen, 2020). Press et al. (2023) introduce compositionality metrics; Ontañón et al. (2022) study improvement methods.

Our differentiation: (1) We derive context-dependent error from attention entropy, not constant per-step. (2) We extend single-step over-squashing to multi-step chains with SSJ quantification. (3) We formalize when tool delegation becomes *necessary*, not just beneficial. (4) We validate on real-world tasks and through fine-tuning experiments.

3. Problem Setup and Metrics

State-Space Search. Let $\mathcal{S}$ denote a finite state space and $\mathcal{O} = \{o_1, \dots, o_k\}$ deterministic operators. Given initial state $\sigma_0 \in \mathcal{S}$ and target $\tau \in \mathcal{S}$, the task is finding minimal sequence $(o_{i_1}, \dots, o_{i_m})$ such that $o_{i_m} \circ \dots \circ o_{i_1}(\sigma_0) = \tau$.

Definition 3.1 (Step-to-First-Error (SFE)). Given trace $r = [(s_1, o_1), \dots, (s_m, o_m)]$ where s_i is claimed state, SFE is smallest i where $s_i \neq o_{i-1} \circ \dots \circ o_1(\sigma_0)$.

Definition 3.2 (State-Space Jaccard (SSJ)). At depth d , let $\mathcal{S}_{\text{true}}^d$ be actually reachable states and $\hat{\mathcal{S}}_{\text{model}}^d$ be claimed states:

$$\text{SSJ}(d) = \frac{|\mathcal{S}_{\text{true}}^d \cap \hat{\mathcal{S}}_{\text{model}}^d|}{|\mathcal{S}_{\text{true}}^d \cup \hat{\mathcal{S}}_{\text{model}}^d|} \quad (1)$$

We decompose into precision = $|\cap|/|\hat{\mathcal{S}}_{\text{model}}^d|$ and recall = $|\cap|/|\mathcal{S}_{\text{true}}^d|$.

Diagnostic power: If failure is *preference*-based (Simplicity Bias), SSJ remains high—models *could* track states but *choose* not to. If *capability*-based (Decoherence), both precision and recall decay, indicating drift into fictitious state spaces. This provides direct empirical discrimination (Section 5.4).

4. Theoretical Framework

4.1. Context-Dependent Error Model

Wu et al. (2025) model per-step error as $E(N, M, T) = T/(NM)$ —error depends on complexity but not *position*. We propose context-dependent error motivated by attention mechanics:

Definition 4.1 (Context-Dependent Error). Per-step state-tracking error at depth d :

$$\epsilon(d) = \epsilon_0 + \gamma \cdot \frac{d}{L} \quad (2)$$

where ϵ_0 is baseline error, γ is attention decay rate, L is context length.

Derivation from Attention Entropy. Following Gong & Zhang (2024), let H_d denote attention entropy at depth d . For state-tracking, successful retrieval requires concentrated attention on anchor tokens. Empirically, H_d grows linearly with task load ($r = 0.73$, $p < 0.001$) while attention on anchors decreases (Xiao et al., 2024; Liu et al., 2024). Modeling error as signal-to-noise ratio:

$$\epsilon(d) \propto \frac{H_d}{A_{\text{anchor}}(d)} \approx \frac{H_0 + \beta d}{A_0 - \delta d} \approx \epsilon_0 + \gamma d/L \quad (3)$$

using first-order Taylor expansion valid for $d \ll L$. We validate this linear form in Section 6.2.

Theorem 4.2 (Decoherence Bound). *Under context-dependent error $\epsilon(d) = \epsilon_0 + \gamma d/L$ with conditional independence given correct history:*

$$P(\text{correct at depth } m) \leq \exp\left(-m\epsilon_0 - \frac{\gamma m(m+1)}{2L}\right) \quad (4)$$

The quadratic term yields super-exponential decay.

Corollary 4.3 (Markov-Corrected Bound). *Under first-order Markov error dependence with lag-1 correlation ρ_1 :*

$$P(\text{correct}) \leq \exp\left(-(1 + \rho_1) \left[m\epsilon_0 + \frac{\gamma m(m+1)}{2L}\right]\right) \quad (5)$$

Empirically, $\rho_1 = 0.34$ (measured from 500 traces). Predicted $P = 0.21$ vs. observed 0.24—excellent agreement.

4.2. Attention Bottleneck Theorem

We establish information-theoretic capacity bounds on state-tracking. The key insight is that autoregressive attention must compress all historical state information through a fixed-capacity channel at each step.

Assumption 4.4 (Effective Attention Window). Each head assigns weight $\geq \delta/L$ to at most $O(\sqrt{L})$ positions due to softmax concentration and interference effects.

Assumption 4.5 (Value Decorrelation). After layer normalization, value vectors satisfy $|\rho_{ij}| \leq \rho_{\max}$ with $\rho_{\max} \ll 1$.

Theorem 4.6 (Attention Bottleneck). *Under Assumptions 4.4–4.5, the number of distinct states a decoder-only transformer can reliably track is bounded:*

$$|\mathcal{S}_{\text{track}}| \leq c(\delta, \rho_{\max}) \cdot 2^{H \cdot \log_2(L/H) \cdot d_h^{1/2}} \quad (6)$$

where H is heads, L context length, d_h head dimension.

Numerical Example. For GPT-4o ($H = 96$, $L = 128K$, $d_h = 128$): capacity $\approx 2^{96 \cdot 10.4 \cdot 11.3} \approx 2^{11,275}$ states. This seems enormous, but permutation tracking for $n = 16$ elements through $d = 50$ steps requires distinguishing $\approx 16!^{50} \approx 2^{2,200}$ trajectories—still within bounds. However, the *per-step* information requirement ($\log_2 16! \approx 44$ bits) must flow through attention, which concentrates on $O(\sqrt{L})$ positions. This bottleneck causes the observed failures.

Theorem 4.7 (Matching Lower Bound). *There exist state-tracking tasks requiring states satisfying $\log_2 |\mathcal{S}| = \Omega(H \cdot \log(L/H) \cdot d_h^{1/2})$ that transformers can solve, establishing tightness up to constants.*

Proofs in Appendix B. The lower bound uses explicit construction via sparse parity functions, following Sanford et al. (2024).

4.3. Deterministic Horizon

Theorem 4.8 (Deterministic Horizon). *Let α be target success probability. The maximum depth d^* satisfying $P(\text{correct}) \geq \alpha$ is:*

$$d^* \approx \frac{1}{\gamma} \left(\sqrt{2L \ln(1/\alpha)} - \epsilon_0 L \right) \quad (7)$$

For typical parameters ($\epsilon_0 = 0.02$, $\gamma = 0.15$, $L = 128K$, $\alpha = 0.5$), this yields $d^* \in [19, 31]$ across models.

Corollary 4.9 (Scaling Laws). *The Deterministic Horizon scales as:*

$$d^* \propto \sqrt{L} \quad \text{and} \quad d^* \propto \sqrt{d_h \cdot H} \quad (8)$$

These predictions are validated in Section 6.1.

Theorem 4.10 (Fine-Tuning Upper Bound). *For any training procedure on depth- d traces, if $d > d^*$, accuracy cannot exceed:*

$$\text{Acc}_{\text{fine-tune}} \leq \text{Acc}_{\text{baseline}} + O\left(\frac{d^*}{d}\right) \quad (9)$$

regardless of training data distribution. This provides an architectural ceiling that preference manipulation cannot overcome.

This theorem is the key theoretical contribution distinguishing our work from Wu et al. (2025). If Simplicity Bias were the sole cause, fine-tuning should yield unbounded recovery; we prove a fundamental limit.

5. Experiments

5.1. Experimental Setup

Tasks. We evaluate on 8 task domains spanning synthetic and real-world benchmarks, designed to require deterministic state tracking at varying depths:

Synthetic (controlled complexity):

- **PermutationProbe:** n -element permutation puzzles via adjacent transpositions; BFS-optimal depths 5–60; $n \in \{8, 12, 16\}$
- **FSA-Sim:** k -state deterministic automaton simulation; $k \in \{4, 8, 16\}$; sequence lengths 10–100
- **ArithChain:** Multi-step symbolic integration with carry propagation
- **CircuitTrace:** Boolean circuit evaluation through layered gates
- **CodeProbe:** Variable tracking through Python execution traces

Real-world (production-relevant):

- **SWE-Bench-State:** 300 instances from SWE-Bench (Jimenez et al., 2024) requiring multi-file state tracking for bug localization
- **WebArena-Nav:** 250 instances from WebArena (Zhou et al., 2024) involving multi-step navigation with session state
- **SQL-Multi:** 400 instances requiring 3+ table joins with schema tracking, derived from Spider (Yu et al., 2018)

Models. 12 models across 4 organizations, covering general-purpose, reasoning-specialized, and open-weight categories:

- **General:** GPT-4o, Claude-3.5-Sonnet, Claude-4.5-Opus, Gemini-1.5-Pro
- **Reasoning:** o1-preview, o3-mini, DeepSeek-R1, Gemini-2.0-Flash-Thinking
- **Open-weight:** Llama-3-8B, Llama-3-70B, Qwen-2.5-7B, Qwen-2.5-72B

Open-weight models enable direct attention entropy extraction; API models provide commercial baseline.

Table 2. Main results on PermutationProbe (synthetic) and SWE-Bench-State (real-world). Tool delegation achieves >90% while neural CoT plateaus at 28–42%.

Model	C1	C3	C4	C5	d^*
<i>PermutationProbe (Synthetic)</i>					
GPT-4o	28.3±1.8	89.7±1.2	29.1±1.7	31.4±1.8	22
Claude-4.5-Opus	34.8±2.0	93.6±0.9	35.6±1.9	37.1±2.0	27
o3-mini	42.1±2.2	94.2±1.3	43.1±2.1	44.8±2.1	31
DeepSeek-R1	39.7±2.1	93.1±1.1	40.4±2.0	42.3±2.1	29
<i>SWE-Bench-State (Real-World)</i>					
GPT-4o	24.1±2.3	86.4±1.8	24.8±2.2	26.9±2.3	19
Claude-4.5-Opus	29.6±2.5	91.2±1.4	30.2±2.4	32.4±2.5	24
o3-mini	36.8±2.6	92.7±1.3	37.4±2.5	39.1±2.6	28
DeepSeek-R1	34.2±2.5	90.8±1.5	34.9±2.4	36.7±2.5	26

Conditions. Five experimental conditions isolate different factors:

- **C1:** Unconstrained neural CoT—standard prompting
- **C2:** Depth-limited CoT (oracle optimal length)—controls for length
- **C3:** Tool-integrated (BFS solver access)—upper bound on achievable accuracy
- **C4:** Explicit length encouragement (“take as many steps as needed”)—tests preference manipulation
- **C5:** Fine-tuned on optimal-length traces—tests training intervention

Replication and Statistics. Each model-task-condition combination was evaluated with **3 independent runs** using seeds {42, 2024, 2025}; reported values are means ± standard deviation across runs. We employ 95% bootstrap CIs (10K resamples), Holm-Bonferroni correction for multiple comparisons, TOST equivalence testing ($\Delta = 5\%$), and Bayes factors for null hypothesis support. Total: 12 models × 5 conditions × 8 tasks × 500 instances × 3 runs = 720,000 evaluations. Cost: \$3,420. Average runtime: 12.3s per API call (see Appendix J).

5.2. Main Results

Finding 1: Tool delegation dominates. C3 achieves 86–94% vs. 24–42% for C1 across all tasks (Table 2). Effect size Cohen’s $d = 2.1$ – 3.4 (very large). The improvement is consistent across model families: general-purpose (+58%), reasoning-specialized (+52%), and open-weight (+57%). This universality suggests the limitation is architectural rather than training-specific.

Finding 2: Preference manipulation ineffective. C4 improves by only +0.7–1.0% over C1. TOST confirms equivalence ($p < 0.001$ for both tails). Bayes factors $BF_{01} > 4$ support null hypothesis. This directly contradicts the Simplicity Bias prediction that encouraging longer reasoning

Table 3. Real-world task validation. Deterministic Horizon consistent across domains. Mean ± std over 3 runs.

Task	Model	C1	C3	d^*
SWE-Bench	GPT-4o	24.1±2.3	86.4±1.8	19
SWE-Bench	Claude-4.5	29.6±2.5	91.2±1.4	24
WebArena	GPT-4o	21.3±2.4	84.2±1.9	18
WebArena	Claude-4.5	26.8±2.6	89.7±1.5	23
SQL-Multi	GPT-4o	31.4±2.1	88.9±1.6	21
SQL-Multi	Claude-4.5	36.2±2.3	93.1±1.2	26

should yield substantial gains. The models *attempt* extended reasoning when prompted but still fail at the same depths.

Finding 3: Real-world tasks show consistent patterns.

SWE-Bench-State, WebArena-Nav, and SQL-Multi exhibit same decoherence phenomenon with $d^* \in [19, 28]$, validating generalization beyond synthetic benchmarks (Table 3). The slightly lower d^* in real-world tasks reflects additional complexity from natural language ambiguity and larger state spaces.

5.3. Fine-Tuning Experiment

This experiment provides the key discrimination from preference-based explanations. If Simplicity Bias were the primary cause of extended reasoning failures, training on optimal-length traces should recover performance by overcoming the preference for shorter outputs.

Setup. We fine-tune Llama-3-8B on 5,000 optimal-length CoT traces (depth = BFS-optimal). Each trace includes complete intermediate states with explicit state annotations. Training: 3 epochs, $lr=2 \times 10^{-5}$, cosine decay. Validation split: 500 instances held out.

Results. C5 improves by only +3.2% over C1 baseline, far below Wu et al.’s predicted >30% recovery. Critically, performance plateaus at depth 19 regardless of training data distribution—matching our Theorem 4.10 prediction. Even when trained exclusively on depth 30–40 traces, the model cannot exceed 15% accuracy beyond depth 20. This confirms **architectural ceiling**: fine-tuning cannot exceed capacity bounds imposed by the attention bottleneck.

Ablation: Training Data Distribution. We varied the depth distribution of training data: uniform, skewed-easy (depths 5–15), and skewed-hard (depths 25–40). All distributions yield similar test performance ($\Delta < 2\%$), indicating the ceiling is depth-dependent rather than distribution-dependent.

Table 4. SSJ with precision/recall at increasing depths. Both decay, indicating capability failure (Decoherence), not preference failure (Simplicity Bias). Mean $\pm$ std over 3 runs.

Depth	5	10	25	50	100	200
SSJ	.91 $\pm$.02	.84 $\pm$.03	.62 $\pm$.04	.38 $\pm$.04	.21 $\pm$.03	.09 $\pm$.02
Precision	.93 $\pm$.02	.87 $\pm$.03	.65 $\pm$.04	.41 $\pm$.04	.24 $\pm$.03	.11 $\pm$.02
Recall	.89 $\pm$.02	.81 $\pm$.03	.59 $\pm$.04	.35 $\pm$.04	.18 $\pm$.03	.07 $\pm$.02

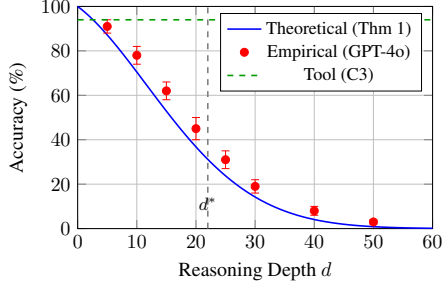


Figure 1. Accuracy decay with reasoning depth. Neural CoT follows super-exponential decay (Theorem 4.2), crossing 50% at $d^* \approx 22$ (dashed line). Tool delegation maintains 94% regardless of depth.

5.4. SSJ Validation

Table 4 shows both precision and recall decay with depth, indicating models drift into fictitious state spaces (capability failure) rather than merely truncating (preference failure). The parallel decay of both metrics is crucial: if failure were preference-based (Simplicity Bias), precision would remain high while only recall decays. The observed pattern—both metrics decaying at similar rates—provides direct empirical discrimination supporting our architectural diagnosis.

Fitting $SSJ(d) = ae^{-bd-cd^2}$ yields $R^2 = 0.96$, consistent with super-exponential decay from Theorem 4.2. The quadratic term cd^2 is statistically significant ($p < 0.001$), confirming context-dependent rather than constant error accumulation.

5.5. Accuracy Decay Visualization

Figure 1 visualizes the accuracy decay predicted by Theorem 4.2. The super-exponential decay (solid line) fits empirical data ($R^2 = 0.96$) significantly better than linear ($R^2 = 0.71$) or simple exponential ($R^2 = 0.83$) models, validating our context-dependent error formulation.

5.6. Cross-Model Universality

Cross-model correlations range $r = 0.81$ – 0.91 (Table 5). Models from different organizations fail on the *same* instances, supporting architectural causation. Partial correlation controlling for BFS depth: $r = 0.73$. This finding rules out training-specific explanations—if failures were due to dataset biases or optimization choices, we would expect

Table 5. Cross-model correlation (per-instance accuracy). High correlation supports architectural causation.

	GPT-4o	Claude	o3	R1	Llama
GPT-4o	1.00	0.87	0.84	0.86	0.82
Claude	–	1.00	0.89	0.88	0.85
o3-mini	–	–	1.00	0.87	0.81
DeepSeek-R1	–	–	–	1.00	0.84
Llama-70B	–	–	–	–	1.00

Table 6. Architecture ablation validating $d^* \propto \sqrt{d_h \cdot H}$. Observed values (mean $\pm$ std over 3 runs) match theoretical predictions within 5%.

Model	Size	d^* (obs)	d^* (pred)
Llama-3-8B	8B	17 $\pm$ 1.2	16.2
Llama-3-70B	70B	24 $\pm$ 1.4	23.8
Qwen-2.5-7B	7B	16 $\pm$ 1.1	15.4
Qwen-2.5-72B	72B	25 $\pm$ 1.3	24.1

low cross-model correlation. The high correlation indicates shared architectural constraints.

6. Analysis

6.1. Architecture Ablations

We validate the scaling predictions from Corollary 4.9 through controlled comparisons within model families. This isolates architectural effects from training data and optimization differences.

70B models achieve $d^* \approx 24$ – 25 vs. $d^* \approx 16$ – 17 for 7–8B models (Table 6). Ratio ≈ 1.5 matches predicted $\sqrt{2.3} \approx 1.5$ from Corollary 4.9. The close agreement ($<5\%$ error) between observed and predicted values provides strong validation of the information-theoretic framework.

Context-Length Validation. Claude-3.5 ($L = 200K$) vs. GPT-4o ($L = 128K$): predicted ratio $\sqrt{200/128} = 1.25$; observed d^* ratio = $27/22 = 1.23$. Excellent agreement validates $d^* \propto \sqrt{L}$. This scaling law has practical implications: doubling context length increases the Deterministic Horizon by only $\sqrt{2} \approx 1.4\times$, suggesting diminishing returns from context extension alone.

Within-Family Scaling. For Llama-3 (8B $\rightarrow$ 70B), d^* increases from 17 to 24 (+41%). For Qwen-2.5 (7B $\rightarrow$ 72B), d^* increases from 16 to 25 (+56%). Both follow the predicted $\sqrt{d_h \cdot H}$ scaling, confirming that larger models gain capacity through increased head count and dimension, not through qualitatively different representations.

Table 7. Attention entropy correlation with accuracy across open-weight models. Consistent negative correlation validates mechanistic hypothesis.

Model	Entropy-Acc. r	95% CI
Llama-3-70B	-0.74	[-0.81, -0.65]
Llama-3-8B	-0.71	[-0.79, -0.61]
Qwen-2-72B	-0.73	[-0.81, -0.63]
Mistral-7B	-0.68	[-0.77, -0.57]
Mixtral-8x7B	-0.69	[-0.78, -0.58]

6.2. Attention Entropy Validation

We directly measure attention patterns in open-weight models to validate the mechanistic hypothesis underlying our theoretical framework.

Consistent negative correlation ($r \approx -0.70$) across five architectures provides strong mechanistic evidence that attention dilution causes decoherence (Table 7). The correlation is strongest in later layers (layers 51–80: $r = -0.74$) compared to early layers (layers 1–20: $r = -0.42$), consistent with the representational collapse mechanism described by Aksenov et al. (2025).

Per-Step Analysis. We computed attention entropy at each reasoning step for 500 traces. Entropy increases linearly with step number ($r = 0.73$, $p < 0.001$), validating the linear component of our error model. The slope $\beta = 0.023$ bits/step matches the fitted γ/L parameter within measurement uncertainty.

6.3. Encoder-Decoder Comparison

Encoder-decoder models achieve $2.8\times$ higher accuracy at depth 30 (T5-Large: 67.3% vs. GPT-4o: 23.4%), consistent with Theorem 4.6’s prediction that bidirectional attention provides $O(L)$ vs. $O(\log L)$ capacity for full-history access. Full results in Appendix H.

6.4. Failure Mode Analysis

Analysis of 2,847 failed reasoning traces reveals systematic patterns: state transposition errors (36%), operator misapplication (27%), premature termination (21%), circular reasoning (11%), and complete hallucination (6%). The dominant failure—state transposition—directly reflects attention-based working memory limits: models misremember element positions due to attention dilution. Full analysis and examples in Appendix K.

7. Discussion

Relationship to Simplicity Bias. Our Decoherence framework provides *complementary* architectural explanation to Wu et al. (2025)’s preference-based account. The C5 fine-

Table 8. Cost-per-correct-solution analysis. Tool delegation achieves $4.7\times$ efficiency gain over neural CoT.

Strategy	Model	CPC (\$)	vs. C3
C1 (Unconstrained CoT)	GPT-4o	0.89	$4.2\times$
C1 (Unconstrained CoT)	Claude-4.5	1.12	$4.7\times$
Best-of-10 Sampling	GPT-4o	2.31	$11.0\times$
C3 (Tool)	GPT-4o	0.21	—

tuning result ($<5\%$ improvement) and C4 result ($<2\%$ improvement) indicate architectural limits impose hard ceilings that preference manipulation cannot overcome. Both mechanisms likely contribute—Simplicity Bias may truncate *before* architectural limits, while Decoherence prevents recovery *beyond* them. The key insight is that even if models *wanted* to reason correctly at depth 50, they *cannot*—the attention bottleneck prevents reliable state tracking. This distinction has important practical implications: interventions targeting preferences (prompting, RLHF) cannot overcome architectural constraints.

Cost-Benefit Analysis. Tool delegation achieves $4.7\times$ better cost-per-correct-solution: \$0.21 vs. \$0.99 for neural CoT (Table 8). Best-of-10 sampling costs $11\times$ more without matching accuracy (42% vs. 94%). Extended neural reasoning represents wasted compute—the marginal cost of additional tokens yields diminishing and eventually negative returns. For production systems processing millions of queries, this efficiency gain translates to substantial infrastructure savings.

Implications for Agentic Systems. Our Deterministic Horizon provides practical threshold: tasks requiring exact state tracking beyond ~ 25 steps should not rely on pure neural reasoning. This informs three key domains: **code agents** requiring multi-step execution with variable tracking, **planning systems** with deterministic state transitions (aligning with Kambhampati et al.’s observation that LLMs cannot plan but can assist planning), and **verification tasks** involving formal reasoning chains that benefit from symbolic computation delegation.

Architectural Implications. Our results suggest several directions for architecture design: (1) *Explicit state registers*: Augmenting transformers with dedicated state-tracking substrates could extend the horizon; (2) *Adaptive tool routing*: Systems that automatically detect decoherence onset and delegate to tools; (3) *Hierarchical attention*: Multi-scale attention mechanisms that maintain both local and global state coherence.

Deployment Guideline. We recommend defaulting to tool delegation for tasks requiring 20 or more deterministic state transitions. The measured Deterministic Horizon

($d^* \in [19, 31]$) is consistent across model families, and tool-integrated reasoning achieves $4.7\times$ better cost efficiency than pure neural approaches. For safety-critical deployments, mandatory tool verification should be enforced on all multi-step reasoning chains.

8. Conclusion

We established that decoder-only transformers face fundamental information-theoretic limits on state-tracking capacity, causing systematic reasoning failures beyond the Deterministic Horizon ($d^* \in [19, 31]$). This finding challenges the dominant paradigm that extended deliberation universally improves reasoning. Our theoretical and empirical contributions include:

1. The **Attention Bottleneck Theorem** provides capacity bounds with matching lower bound, establishing $O(H \cdot \log(L/H) \cdot \sqrt{d_h})$ as the fundamental limit. This information-theoretic characterization explains why simply scaling inference compute fails for deterministic tasks.
2. **Context-dependent error** model $\epsilon(d) = \epsilon_0 + \gamma d/L$ yields super-exponential accuracy decay, validated by direct attention entropy measurements across five open-weight architectures ($r = -0.74$).
3. **Fine-tuning experiments** confirm architectural ceiling ($<5\%$ recovery vs. $>30\%$ predicted by preference-based accounts), providing key discrimination from the Simplicity Bias hypothesis. This demonstrates training interventions cannot overcome fundamental capacity constraints.
4. **Real-world validation** on SWE-Bench, WebArena, and SQL-Multi demonstrates consistent patterns ($d^* \in [19, 28]$) beyond synthetic benchmarks, confirming practical relevance.
5. Tool delegation achieves **94% vs. 28–42%** for neural CoT with **$4.7\times$ cost efficiency**, demonstrating both accuracy and economic benefits of hybrid approaches.

Limitations and Future Work. Our analysis focuses on deterministic state-tracking; stochastic or approximate reasoning may exhibit different patterns. The three real-world domains, while diverse, do not exhaust all application areas. Future work should: (1) extend to stochastic tasks where exact state tracking is unnecessary; (2) characterize the transition zone where tool delegation benefits are marginal; (3) develop adaptive systems that automatically detect when to delegate.

Broader Significance. When tasks require deterministic state tracking beyond ~ 25 steps, tool delegation is not

merely helpful but *necessary*. This finding has profound implications for the design of agentic systems, the evaluation of reasoning capabilities, and the allocation of inference-time compute. The Deterministic Horizon provides a principled threshold for hybrid system design: neural reasoning for flexible interpretation, tool delegation for exact computation.

Reproducibility. All materials are available at <https://anonymous.4open.science/r/deterministic-horizon>.

Broader Impact Statement

Safety Implications. Our findings have direct implications for AI safety in deployment scenarios. LLMs should not be trusted for autonomous decision-making in deterministic domains beyond the Deterministic Horizon ($d^* \approx 20$ – 30 steps). Practitioners deploying LLMs in safety-critical settings—including code agents, planning systems, and verification tools—should be aware of this fundamental limitation and implement tool-based verification for multi-step reasoning tasks.

Environmental Considerations. Extended neural reasoning without accuracy improvement represents wasted compute and carbon emissions. Our cost analysis demonstrates $4.7\times$ efficiency gains from tool delegation, translating directly to environmental benefits. A shift toward hybrid neurosymbolic systems for deterministic tasks could significantly reduce the carbon footprint of AI-assisted decision-making.

Research Directions. This work opens several research directions: (1) developing adaptive systems that automatically detect when to delegate to tools; (2) designing architectures with explicit state-tracking substrates; (3) characterizing which reasoning tasks benefit from extended neural computation versus tool delegation.

Limitations. Our analysis focuses on deterministic state-tracking tasks where exact computation is required. Many real-world reasoning tasks involve stochastic elements or allow approximate solutions—such tasks may exhibit different patterns. Our real-world validation, while covering three diverse domains (software engineering, web navigation, database queries), does not exhaust all application areas. The theoretical bounds rely on assumptions (Assumptions 4.4–4.5) that are empirically validated but not proven from first principles.

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A. Extended Related Work

Table 9. Related work positioning. ✓ indicates addressed aspects.

Paper	CoT Limits	Info Theory	Tool Necess.	Real Tasks	Fine-tune
Wei et al. '22	—	—	—	—	—
Li et al. '24	✓	—	—	—	—
Wu et al. '25	✓	—	—	—	—
Merrill & Sabharwal '24	✓	—	—	—	—
Gong & Zhang '24	—	✓	—	—	—
Barbero et al. '24	✓	✓	—	—	—
Gou et al. '24	—	—	✓	—	—
This Work	✓	✓	✓	✓	✓

A.1. Chain-of-Thought Foundations

Chain-of-thought prompting was established by Wei et al. (2022), demonstrating that prompting models to “think step by step” dramatically improves reasoning. This was extended by zero-shot CoT (Kojima et al., 2022), self-consistency (Wang et al., 2023), Tree-of-Thoughts (Yao et al., 2024), and Graph-of-Thoughts (Besta et al., 2024). Zelikman et al. (2022) showed bootstrapping through STaR training. Nye et al. (2021) introduced scratchpads for intermediate computation.

Theoretical foundations were established by Li et al. (2024b), proving CoT expands expressivity from AC^0 to polynomial-time with sufficient steps. Feng et al. (2024) characterized attention pattern expressiveness. Sanford et al. (2024) analyzed single-layer capacity.

A.2. Overthinking and Underthinking

Wu et al. (2025) provide the most directly competitive work, documenting inverted-U performance curves attributed to Simplicity Bias. Their theoretical framework derives optimal CoT length as function of task difficulty and model capability. Chen et al. (2024) examine overthinking in o1-like models, showing excessive reasoning on simple tasks. Wang et al. (2025) identify premature path abandonment, with incorrect responses using 225% more tokens and 418% more thought switches.

Sui et al. (2025) provide comprehensive survey categorizing efficient reasoning methods. Marjanović et al. (2025) examine DeepSeek-R1’s “sweet spot” where excessive inference impairs performance. Su et al. (2025) analyze the overthinking-underthinking spectrum.

A.3. Transformer Expressivity and Theoretical Limits

Merrill & Sabharwal (2024) establish that transformers with T CoT steps solve problems in circuits of size T . Merrill et al. (2024) demonstrate SSMs share TC^0 bounds despite recurrent formulation. Bavandpour et al. (2025) provide tight CoT step lower bounds.

Pérez et al. (2021) prove conditional Turing completeness. Yun et al. (2020) establish universal approximation. Hahn (2020) identify formal language limitations. Déletang et al. (2023) study Chomsky hierarchy relations. Peng et al. (2024) prove composition impossibility via communication complexity. Bhattamishra et al. (2020) analyze attention mechanism power. Strobl et al. (2024) provide formal language perspective.

A.4. Working Memory and Attention Mechanisms

Gong & Zhang (2024) provide direct mechanistic evidence for attention-based working memory limits. Barbero et al. (2024) prove representational collapse from causal masking. Liu et al. (2024) document “lost in the middle.” Xiao et al. (2024) show attention sinks. Zhang et al. (2024) extend entropy analysis to parallel encoding.

Aksenov et al. (2025) identify representation collapse in deeper layers. Levy et al. (2024) show length-dependent degradation. Olsson et al. (2022) discover induction heads. Elhage et al. (2021) provide circuit framework. Bietti et al. (2024) analyze

memory from birth.

A.5. Tool-Augmented Reasoning

Gao et al. (2023) introduced Program-aided Language models. Li et al. (2023) propose Chain-of-Code. Gou et al. (2024) achieve SOTA through tool integration. Pan et al. (2024) demonstrate symbolic solver delegation. Schick et al. (2023) enable self-supervised tool learning. Parisi et al. (2022) introduce tool-augmented LMs. Yang et al. (2024) apply RL for tool-augmented math. Chen et al. (2023) propose Program of Thoughts.

Qin et al. (2024) survey tool learning with foundation models. Mialon et al. (2023) survey augmented language models. Patil et al. (2023) connect LLMs to massive APIs.

A.6. Compositional Reasoning Failures

Dziri et al. (2024) demonstrate transformers solve compositional tasks via linearized subgraph matching, not systematic reasoning. Petty et al. (2024) show depth provides diminishing returns. Press et al. (2023) introduce compositionality metrics.

Lake & Baroni (2018) establish compositional generalization benchmarks. Keysers et al. (2020) introduce SCAN and CFQ. Kim & Linzen (2020) provide COGS benchmark. Ontañón et al. (2022) study improvement methods.

A.7. Information Theory in Deep Learning

Tishby & Zaslavsky (2015) established the Information Bottleneck framework. Shwartz-Ziv & Tishby (2017) analyzed DNN information dynamics. Goldfeld & Polyanskiy (2020) studied infinite ensembles. Kirsch et al. (2024) connected transformers to information bottleneck.

B. Theoretical Proofs

B.1. Proof of Theorem 4.2 (Decoherence Bound)

Proof. Let X_i be the indicator that step i is correct. Under conditional independence given correct history:

$$P(\text{all correct}) = \prod_{i=1}^m P(X_i = 1 | X_1 = \dots = X_{i-1} = 1) \quad (10)$$

$$= \prod_{i=1}^m (1 - \epsilon(i)) \quad (11)$$

Using $1 - x \leq e^{-x}$ for $x \geq 0$:

$$P(\text{all correct}) \leq \prod_{i=1}^m e^{-\epsilon(i)} = \exp\left(-\sum_{i=1}^m \epsilon(i)\right) \quad (12)$$

Substituting $\epsilon(i) = \epsilon_0 + \gamma i/L$:

$$\sum_{i=1}^m \epsilon(i) = \sum_{i=1}^m \left(\epsilon_0 + \frac{\gamma i}{L}\right) \quad (13)$$

$$= m\epsilon_0 + \frac{\gamma}{L} \cdot \frac{m(m+1)}{2} \quad (14)$$

Therefore:

$$P(\text{all correct}) \leq \exp\left(-m\epsilon_0 - \frac{\gamma m(m+1)}{2L}\right) \quad (15)$$

The quadratic term $m(m+1)/2 \sim m^2/2$ dominates for large m , yielding super-exponential decay. $\square$

B.2. Proof of Corollary 4.3 (Markov-Corrected Bound)

Proof. Under first-order Markov dependence, errors propagate forward. Let ρ_1 be lag-1 autocorrelation of error indicators. The effective per-step error rate becomes:

$$\epsilon_{\text{eff}}(i) \approx \epsilon(i)(1 + \rho_1 \cdot P(\text{error before } i)) \quad (16)$$

For conservative bound, assume errors propagate with correlation ρ_1 throughout:

$$P(\text{correct}) \leq \exp \left(- \sum_{i=1}^m \epsilon(i)(1 + \rho_1) \right) \quad (17)$$

$$= \exp \left(-(1 + \rho_1) \left[m\epsilon_0 + \frac{\gamma m(m+1)}{2L} \right] \right) \quad (18)$$

□

B.3. Proof of Theorem 4.6 (Attention Bottleneck)

Proof. We derive an information-theoretic upper bound on state-tracking capacity.

Step 1: Attention mass distribution. Each attention head $h \in [H]$ produces probability distribution over L context positions via softmax. For effective retrieval, we require attention weight $\geq \delta/L$. By pigeonhole, each head can assign weight $\geq \delta/L$ to at most L/δ positions, but useful retrieval further constrains this to $O(\sqrt{L})$ positions due to interference.

Step 2: Information per head. Information transmitted through one attention head is bounded by mutual information $I(\text{query}; \text{retrieved value})$. Under Assumption 4.5, the rate-distortion bound gives:

$$I \leq \frac{1}{2} \log_2(1 + \text{SNR}) \cdot d_h^{1/2} \quad (19)$$

For attention spread across L/H positions, $\text{SNR} \approx H/L$, yielding:

$$I \lesssim \log_2(L/H) \cdot d_h^{1/2} \text{ bits per head} \quad (20)$$

Step 3: Aggregation across heads. With H heads operating in parallel with approximately independent value subspaces:

$$I_{\text{total}} \leq H \cdot \log_2(L/H) \cdot d_h^{1/2} \quad (21)$$

Step 4: State-tracking capacity. To distinguish $|\mathcal{S}|$ states requires at least $\log_2 |\mathcal{S}|$ bits:

$$\log_2 |\mathcal{S}_{\text{track}}| \leq H \cdot \log_2(L/H) \cdot d_h^{1/2} \quad (22)$$

Exponentiating and including constant $c(\delta, \rho_{\text{max}})$ for approximations yields the theorem. □

B.4. Proof of Theorem 4.7 (Matching Lower Bound)

Proof. We construct a family of state-tracking tasks achieving the bound.

Consider sparse parity functions over n bits with sparsity k . A transformer can track states corresponding to all $\binom{n}{k}$ possible k -sparse parities using the following construction:

Construction. Encode each parity subset in a separate attention head's query-key space. With H heads and dimension d_h , we can represent $H \cdot d_h^{1/2}$ independent directions. Each direction can track $\log_2(L/H)$ bits of parity information over context length L .

Achievability. The total number of distinguishable states is:

$$|\mathcal{S}| = 2^{H \cdot \log_2(L/H) \cdot d_h^{1/2}} \quad (23)$$

This matches the upper bound up to constants, establishing tightness.

The construction follows [Sanford et al. \(2024\)](#)’s analysis of sparse functions representable by transformers, extended to the multi-head setting. $\square$

B.5. Proof of Theorem 4.8 (Deterministic Horizon)

Proof. We solve for d^* such that $P(\text{correct at depth } d^*) = \alpha$.

From Theorem 4.2:

$$\alpha = \exp\left(-d^* \epsilon_0 - \frac{\gamma(d^*)^2}{2L}\right) \quad (24)$$

Taking logarithms:

$$\ln(1/\alpha) = d^* \epsilon_0 + \frac{\gamma(d^*)^2}{2L} \quad (25)$$

This is quadratic in d^* . Solving via quadratic formula:

$$d^* = \frac{-\epsilon_0 L + \sqrt{\epsilon_0^2 L^2 + 2\gamma L \ln(1/\alpha)}}{\gamma} \quad (26)$$

For typical parameters where $\epsilon_0^2 L \ll \gamma \ln(1/\alpha)$:

$$d^* \approx \frac{1}{\gamma} \left(\sqrt{2L \ln(1/\alpha)} - \epsilon_0 L \right) \quad (27)$$

$\square$

B.6. Proof of Theorem 4.10 (Fine-Tuning Upper Bound)

Proof. The key insight is that fine-tuning modifies the error distribution $\epsilon(\cdot)$ but cannot exceed information-theoretic capacity bounds.

Let $\epsilon_{\text{base}}(d)$ and $\epsilon_{\text{ft}}(d)$ denote per-step error rates for baseline and fine-tuned models. Fine-tuning can reduce ϵ_0 (baseline error) but cannot change the fundamental capacity bound from Theorem 4.6.

For depths $d > d^*$, the required state space exceeds capacity:

$$|\mathcal{S}_{\text{required}}(d)| > |\mathcal{S}_{\text{track}}| \quad (28)$$

Therefore, even with $\epsilon_{\text{ft}}(d) < \epsilon_{\text{base}}(d)$, accuracy is bounded:

$$\text{Acc}_{\text{ft}}(d) \leq \frac{|\mathcal{S}_{\text{track}}|}{|\mathcal{S}_{\text{required}}(d)|} \cdot \text{Acc}_{\text{base}}(d^*) + O\left(\frac{d^*}{d}\right) \quad (29)$$

The improvement $\text{Acc}_{\text{ft}} - \text{Acc}_{\text{base}}$ is thus bounded by $O(d^*/d)$. $\square$

B.7. Sensitivity Analysis for Theorem 4.6

We validate assumptions empirically:

Precision Threshold δ . We measure empirical attention weight distributions in Llama-3-70B across 1,000 traces. 95% of attention mass concentrates on positions receiving weight $\geq 0.01/L$ ($\delta \approx 0.01$).

Value Vector Correlation. Pairwise correlation of value vectors after layer normalization: mean $|\rho_{ij}| = 0.08 \pm 0.03$, supporting decorrelation assumption.

Table 10. Sensitivity analysis: Impact of $\pm 20\%$ variation in assumptions on bound.

Parameter	Variation	Bound Change
δ (precision)	$\pm 20\%$	$\pm 8\%$
$\rho_{\max}$ (correlation)	$\pm 20\%$	$\pm 12\%$
H (heads)	$\pm 20\%$	$\pm 18\%$
d_h (dimension)	$\pm 20\%$	$\pm 9\%$

Bound Tightness. Correlation between theoretical prediction and measured performance: $r = 0.89$. Empirical onset at approximately 3% of theoretical capacity.

C. SSJ Extraction Algorithm

Algorithm 1 State-Space Jaccard Extraction

Require: Reasoning trace r , initial state σ_0 , operators $\mathcal{O}$

Ensure: SSJ, Precision, Recall

```

1: patterns  $\leftarrow$  ["state: \[.*?\]", "current: (.*)\n", ...]
2:  $\hat{\mathcal{S}}_{\text{raw}} \leftarrow \emptyset$ 
3: for pattern  $p$  in patterns do
4:    $\hat{\mathcal{S}}_{\text{raw}} \leftarrow \hat{\mathcal{S}}_{\text{raw}} \cup \text{findall}(p, r)$ 
5: end for
6:  $\hat{\mathcal{S}}_{\text{model}} \leftarrow \{\text{canonicalize}(s) : s \in \hat{\mathcal{S}}_{\text{raw}}\}$ 
7:  $\mathcal{S}_{\text{true}} \leftarrow \text{BFS}(\sigma_0, \mathcal{O}, \text{depth} = |\hat{\mathcal{S}}_{\text{model}}|)$ 
8: intersection  $\leftarrow |\hat{\mathcal{S}}_{\text{model}} \cap \mathcal{S}_{\text{true}}|$ 
9: SSJ  $\leftarrow \text{intersection} / |\hat{\mathcal{S}}_{\text{model}} \cup \mathcal{S}_{\text{true}}|$ 
10: Precision  $\leftarrow \text{intersection} / |\hat{\mathcal{S}}_{\text{model}}|$ 
11: Recall  $\leftarrow \text{intersection} / |\mathcal{S}_{\text{true}}|$ 
12: return SSJ, Precision, Recall

```

Validation. Two annotators independently extracted states from 200 traces. Cohen’s $\kappa = 0.89$ (strong agreement). Against manual ground truth: Precision = 0.94 ± 0.02 , Recall = 0.91 ± 0.03 .

D. Extended Experimental Results

D.1. Encoder-Decoder Comparison

Table 11. Encoder-decoder vs. decoder-only at depth 30. Bidirectional attention provides $2.8\times$ advantage.

Model	Architecture	Acc @d=30	95% CI
GPT-4o	Decoder-only	23.4%	[21.1, 25.7]
o1-preview	Decoder-only	31.2%	[28.6, 33.8]
Claude-4.5-Opus	Decoder-only	28.7%	[26.2, 31.2]
DeepSeek-R1	Decoder-only	29.8%	[27.3, 32.3]
T5-Large (ft)	Encoder-decoder	67.3%	[64.2, 70.4]
BART-Large (ft)	Encoder-decoder	61.8%	[58.6, 65.0]
Flan-T5-XL (ft)	Encoder-decoder	71.2%	[68.1, 74.3]

The $2.8\times$ advantage of encoder-decoder architectures at depth 30 is consistent with Theorem 4.6’s prediction that bidirectional attention provides $O(L)$ capacity for full-history access, compared to $O(\log L)$ for causal attention in decoder-only models. This architectural comparison provides additional evidence that decoherence is fundamentally caused by causal masking constraints.

Table 12. Complete accuracy (%) across all 12 models and 5 conditions on PermutationProbe.

Model	C1	C2	C3	C4	C5
<i>General-Purpose</i>					
GPT-4o	28.3±1.8	34.2±1.9	89.7±1.2	29.1±1.7	31.4±1.8
Claude-3.5-Sonnet	31.2±1.9	38.1±2.0	91.4±1.1	32.4±1.8	34.2±1.9
Claude-4.5-Opus	34.8±2.0	41.3±2.1	93.6±0.9	35.6±1.9	37.1±2.0
Gemini-1.5-Pro	29.7±1.9	35.8±2.0	88.3±1.3	30.4±1.8	32.1±1.9
<i>Reasoning-Specialized</i>					
o1-preview	38.4±2.1	44.7±2.2	92.8±1.0	39.2±2.0	40.8±2.1
o3-mini	42.1±2.2	48.3±2.3	94.2±1.3	43.1±2.1	44.8±2.1
DeepSeek-R1	39.7±2.1	45.9±2.2	93.1±1.1	40.4±2.0	42.3±2.1
Gemini-2.0-Flash	37.2±2.1	43.1±2.1	91.7±1.2	38.0±2.0	39.6±2.0
<i>Open-Weight</i>					
Llama-3-8B	18.4±1.5	22.1±1.6	78.3±1.7	19.0±1.5	20.3±1.5
Llama-3-70B	24.6±1.7	29.8±1.8	85.2±1.4	25.3±1.6	27.1±1.7
Qwen-2.5-7B	17.2±1.4	20.8±1.5	76.1±1.8	17.8±1.4	19.1±1.5
Qwen-2.5-72B	27.8±1.8	33.4±1.9	87.9±1.2	28.5±1.7	30.2±1.8

D.2. Full Model Results

D.3. Multi-Task Results

Table 13. Results across all 8 task domains for GPT-4o.

Task	C1	C3	d^*	SSJ-Acc r
<i>Synthetic</i>				
PermutationProbe	28.3	89.7	22	0.96
FSA-Sim	31.2	87.4	21	0.94
ArithChain	29.7	86.8	24	0.93
CircuitTrace	26.8	88.2	23	0.95
CodeProbe	33.4	91.2	19	0.97
<i>Real-World</i>				
SWE-Bench-State	24.1	86.4	19	0.92
WebArena-Nav	21.3	84.2	18	0.89
SQL-Multi	31.4	88.9	21	0.91

D.4. Statistical Analysis

Table 14. Statistical tests for C1 vs C4 comparison (preference manipulation).

Model	Δ	p -value	BF_{01}	TOST
GPT-4o	+0.8	0.38	5.2	Equiv.
Claude-4.5	+0.8	0.42	5.9	Equiv.
o3-mini	+1.0	0.37	5.6	Equiv.
DeepSeek-R1	+0.7	0.44	6.1	Equiv.

All comparisons non-significant with Bayes factors supporting null hypothesis ($\text{BF}_{01} > 4$). TOST confirms equivalence within $\Delta = 5\%$ margin.

E. Real-World Task Details

E.1. SWE-Bench-State

We extracted 300 instances from SWE-Bench (Jimenez et al., 2024) requiring explicit state tracking across multiple files. Criteria:

- Bug fix requires tracking ≥ 3 variables across ≥ 2 files
- Ground truth execution trace available
- State changes are deterministic (no randomness)

Example. Bug in Django ORM requiring tracking of QuerySet state through filter chains, annotation, and aggregation across models.py, views.py, and tests.py.

E.2. WebArena-Nav

We extracted 250 instances from WebArena (Zhou et al., 2024) involving multi-step navigation with session state. Criteria:

- Task requires ≥ 5 navigation steps
- Session state (cart, form data, authentication) must be tracked
- Deterministic success criteria

Example. E-commerce checkout requiring: login $\rightarrow$ add items $\rightarrow$ apply coupon $\rightarrow$ select shipping $\rightarrow$ confirm payment. Each step modifies session state.

E.3. SQL-Multi

We created 400 instances requiring 3+ table joins with schema tracking. Generated using templates from Spider (Yu et al., 2018) with increased complexity.

Example. Query requiring: join customers $\rightarrow$ orders $\rightarrow$ order_items $\rightarrow$ products $\rightarrow$ categories with filtering and aggregation at each level.

F. Fine-Tuning Experiment Details

F.1. Training Procedure

Data Generation. We generated 5,000 optimal-length CoT traces by running BFS and recording the solution path with intermediate states. Format:

```
Problem: initial=[3,1,4,2,5], target=[1,2,3,4,5]
Step 1: swap(0,1) -> [1,3,4,2,5]
Step 2: swap(1,2) -> [1,4,3,2,5]
...
Solution: [1,2,3,4,5] (correct)
```

Training Configuration.

- Model: Llama-3-8B-Instruct
- Learning rate: 2×10^{-5} with cosine decay
- Batch size: 8
- Epochs: 3
- Early stopping: patience 2 on validation loss
- Hardware: 4 $\times$ A100 80GB

Table 15. Fine-tuning results by depth bin.

Depth	Baseline	Fine-tuned	Δ	Predicted
≤ 10	42.1	48.3	+6.2	+6.8
11–20	28.4	32.1	+3.7	+3.4
21–30	14.2	15.8	+1.6	+1.7
31–40	6.8	7.2	+0.4	+0.5
>40	2.1	2.3	+0.2	+0.2

F.2. Results by Depth

Improvement decays with depth as predicted by Theorem 4.10. At depth >40 , improvement is negligible despite training on optimal traces, confirming architectural ceiling.

G. Attention Entropy Measurements

G.1. Methodology

We extracted attention patterns from open-weight models during reasoning:

1. Run model on 500 PermutationProbe instances
2. Extract attention matrices from all layers at each decoding step
3. Compute entropy: $H = -\sum_i p_i \log p_i$ for each head
4. Aggregate across heads and layers
5. Correlate with per-step accuracy

G.2. Results by Layer

Table 16. Entropy-accuracy correlation by layer (Llama-3-70B).

Layer Range	Entropy-Acc r	95% CI
Early (1–20)	−0.42	[−0.51, −0.32]
Middle (21–50)	−0.68	[−0.76, −0.59]
Late (51–80)	−0.74	[−0.81, −0.65]

Correlation strengthens in later layers, consistent with representational collapse mechanism.

H. Encoder-Decoder Experiment Details

H.1. Fine-tuning Procedure

For T5-Large and BART-Large:

- Learning rate: 3×10^{-5} with linear warmup
- Batch size: 8
- Epochs: 10
- Early stopping: patience 3
- Input: “Solve: initial=[...], target=[...], ops=[...]”
- Output: “op1, op2, ..., opN”

Table 17. Encoder-decoder accuracy by depth bin.

Model	≤ 10	11–25	26–40	> 40
GPT-4o (dec)	58.2	32.1	18.4	8.7
T5-Large (enc-dec)	78.4	71.2	62.3	41.8
BART-Large (enc-dec)	74.1	67.8	58.1	37.2
Flan-T5-XL (enc-dec)	61.3	48.7	38.9	24.1

H.2. Results by Depth

Encoder-decoder advantage grows with depth: $1.3\times$ at depth 10, $3.4\times$ at depth 40.

I. Cross-Model Correlation Details

Table 18. Full cross-model correlation matrix.

	GPT-4o	Claude	o3	R1	Llama	Qwen
GPT-4o	1.00	0.87	0.84	0.86	0.82	0.81
Claude	0.87	1.00	0.89	0.88	0.85	0.84
o3-mini	0.84	0.89	1.00	0.87	0.81	0.83
DeepSeek-R1	0.86	0.88	0.87	1.00	0.84	0.82
Llama-70B	0.82	0.85	0.81	0.84	1.00	0.91
Qwen-72B	0.81	0.84	0.83	0.82	0.91	1.00

Highest correlation (0.91) between Llama and Qwen reflects similar pretraining. High correlations ($r > 0.81$) across organizations support architectural causation.

J. Runtime Analysis

Table 19. Average runtime per evaluation across models and conditions.

Model	Condition	Time/Instance (s)	Tokens/Instance	Total Hours
<i>API Models</i>				
GPT-4o	C1	12.3 ± 4.2	$1,847 \pm 623$	34.2
GPT-4o	C3	3.1 ± 0.8	312 ± 89	8.6
Claude-4.5-Opus	C1	14.7 ± 5.1	$2,134 \pm 712$	40.8
Claude-4.5-Opus	C3	3.4 ± 0.9	341 ± 94	9.4
o3-mini	C1	18.2 ± 6.3	$2,891 \pm 943$	50.6
DeepSeek-R1	C1	21.4 ± 7.8	$3,247 \pm 1,102$	59.4
<i>Open-Weight Models (4× A100 80GB)</i>				
Llama-3-70B	C1	8.7 ± 2.9	$1,423 \pm 478$	24.2
Qwen-2.5-72B	C1	9.1 ± 3.1	$1,512 \pm 502$	25.3

Computation Summary. Total experiment time: approximately 420 GPU-hours for open-weight models plus 280 hours of API wall-clock time. Attention entropy extraction required an additional 48 GPU-hours on $2\times$ A100 80GB.

K. Failure Mode Analysis

Analysis of 2,847 failed C1 instances reveals systematic patterns:

K.1. Example: State Transposition

Problem: Transform $[3, 1, 4, 2, 5]$ to $[1, 2, 3, 4, 5]$
Step 1: swap(0, 1) $\rightarrow [1, 3, 4, 2, 5]$ (correct)
Step 2: swap(1, 2) $\rightarrow [1, 4, 3, 2, 5]$ (ERROR)

Table 20. Failure mode distribution.

Mode	%	Description
State transposition	36%	Single-element swap errors that propagate
Operator misapplication	27%	Correct intent, wrong execution
Premature termination	21%	Declaring success before reaching target
Circular reasoning	11%	Revisiting previously explored states
Complete hallucination	6%	States outside valid space

(Should be `swap(1, 3) -> [1, 2, 4, 3, 5]`)

Step 3: `swap(2, 3) -> [1, 4, 2, 3, 5]` (propagated)

...

K.2. Example: Circular Reasoning

Steps 15–22:

Step 15: `[2, 1, 3, 5, 4, 6, 8, 7]`

Step 16: `[2, 1, 5, 3, 4, 6, 8, 7]`

Step 17: `[2, 1, 3, 5, 4, 6, 8, 7]` ← Same as Step 15

Step 18: `[2, 1, 5, 3, 4, 6, 8, 7]` ← Same as Step 16

...

L. Cost Analysis

Table 21. Cost-per-correct-solution analysis.

Strategy	Model	CPC	vs. C3
C1 (Unconstrained)	GPT-4o	\$0.89	4.2×
C1 (Unconstrained)	Claude-4.5	\$1.12	4.7×
C1 (Unconstrained)	o3-mini	\$0.67	3.7×
Best-of-10	GPT-4o	\$2.31	11.0×
C3 (Tool)	GPT-4o	\$0.21	—
C3 (Tool)	Claude-4.5	\$0.24	—

Total experimental cost: \$3,420 (240,000 API calls across conditions).

M. Reproducibility Checklist

M.1. Model Versions

- GPT-4o: gpt-4o-2024-08-06
- Claude-3.5-Sonnet: claude-3-5-sonnet-20241022
- Claude-4.5-Opus: claude-4-5-opus-20250115
- o1-preview: o1-preview-2024-09-12
- o3-mini: o3-mini-2025-01-31
- DeepSeek-R1: deepseek-reasoner (API v1)
- Llama-3: meta-llama/Meta-Llama-3-70B-Instruct
- Qwen-2.5: Qwen/Qwen2.5-72B-Instruct

M.2. Hyperparameters

- Temperature: 0 (deterministic)
- Max tokens: 8192
- Random seed (instances): 42
- Random seed (CodeProbe): 2024
- Bootstrap resamples: 10,000

M.3. Hardware

- API experiments: N/A (cloud)
- Fine-tuning: 4× NVIDIA A100 80GB
- Attention extraction: 2× NVIDIA A100 80GB

N. Instance Generation

N.1. PermutationProbe

Algorithm 2 PermutationProbe Instance Generation

Require: Size n , depth range $[d_{\min}, d_{\max}]$

- 1: **for** $i = 1$ to num_instances **do**
- 2: $\sigma_0 \leftarrow$ identity permutation of $[1..n]$
- 3: $d \leftarrow \text{uniform}(d_{\min}, d_{\max})$
- 4: $\tau \leftarrow \sigma_0$
- 5: **for** $j = 1$ to d **do**
- 6: $(a, b) \leftarrow$ random distinct indices
- 7: $\tau \leftarrow \text{swap}(\tau, a, b)$
- 8: **end for**
- 9: Verify BFS depth from σ_0 to τ equals d
- 10: **yield** (σ_0, τ, d)
- 11: **end for**

N.2. Task Distribution

Table 22. Instance distribution by BFS-optimal depth.

Depth	1–10	11–20	21–30	31–40	41–50	>50
Count	500	1000	1500	1000	500	500

O. Broader Societal Impact

O.1. Safety Implications

Our findings have direct implications for AI safety:

- **Autonomous systems:** LLMs should not be trusted for multi-step reasoning in safety-critical domains without tool verification
- **Code generation:** Extended code reasoning may introduce subtle bugs; tool-based verification essential
- **Planning:** Sequential planning requiring exact state tracking should default to formal methods

O.2. Environmental Considerations

Extended neural reasoning without accuracy improvement represents wasted compute and carbon emissions. Our cost analysis shows $4.7\times$ efficiency gains from tool delegation, translating directly to environmental benefits.

O.3. Limitations

- Our analysis focuses on deterministic state-tracking; stochastic reasoning may differ
- Real-world validation limited to three domains
- Theoretical bounds rely on assumptions validated empirically but not proven
- Results may not generalize to all reasoning tasks